

Hand in

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Problem (1). Let $X = [-1, 1] \times \{-1, +1\} \subseteq \mathbb{R}^2$ have the usual metric topology and define a relation $\sim$ on X saying that $(a, -1) \sim (a, 1)$ for all $a \in [-1, 0) \cup (0, 1]$. Let $Y = X / \sim$ have the quotient topology

- Check that $\sim$ is an equivalence relation
- Is Y Hausdorff?
- Does Y have a countable basis for its topology?
- Check up the definition of connectedness in section 23: is Y connected?
- Check up the definition of compact in section 26: is Y compact?

Solution and proof of all. Let

$$q: X \longrightarrow Y = X / \sim$$

denote the quotient map. It is useful to write

$$0_- := q(0, -1), \quad 0_+ := q(0, 1),$$

and, for $a \neq 0$,

$$\bar{a} := q(a, -1) = q(a, 1).$$

Thus, intuitively, Y is the interval $[-1, 1]$ where every point except 0 has been identified between the two copies, while the origin occurs twice.

The relation $\sim$ is an equivalence relation.

More explicitly, for $(a, s), (b, t) \in X$, we may write

$$(a, s) \sim (b, t) \iff a = b \text{ and either } s = t \text{ or } a \neq 0.$$

We check the three conditions.

- **Reflexivity:** For every $(a, s) \in X$, clearly

$$(a, s) \sim (a, s).$$

- **Symmetry:** If $(a, s) \sim (b, t)$, then $a = b$. If $s = t$, symmetry is immediate. Otherwise $a = b \neq 0$, and by the definition of the relation we also have

$$(b, t) \sim (a, s).$$

- **Transitivity:** Suppose

$$(a, s) \sim (b, t) \quad \text{and} \quad (b, t) \sim (c, u).$$

Then necessarily $a = b = c$. If this common value is nonzero, the two copies are identified, so

$$(a, s) \sim (c, u).$$

If the common value is 0, the only equivalences at 0 are the reflexive ones. Hence $s = t = u$, and again

$$(a, s) \sim (c, u).$$

Therefore $\sim$ is an equivalence relation. □